

Figure 3: Hypertable model

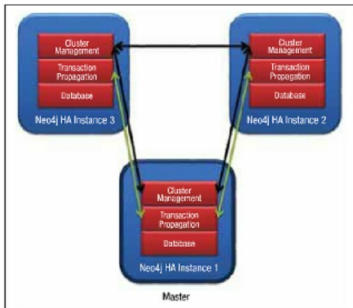


Figure 4: HA in Neo4J

- Clean semantics: It has a consistent database.

Document stores

The main concept of a document store is the document. Document-oriented databases are designed to store, retrieve, and manage document-oriented structures (like XML files—XML data sources leverage XQuery), or semi-structured data (like text files—text documents are indexed and facilitate keyword search-like retrieval). Each document-oriented database encapsulates and encodes data in some standard format or encodings such as XML, YAML, or JSON as well as binary forms such as BSON, PDF, and Microsoft Office documents.

Apache CouchDB

Apache CouchDB is an open source NoSQL database. CouchDB uses JSON (JavaScript Object Notation, lightweight data-interchange format) to store data.

JavaScript is used as its query language. CouchDB is published via the Apache Software Foundation in 2008. In CouchDB, each database is a collection of independent documents. Each document manages its own data and metadata (self-contained schema). CouchDB is ideal in situations where a network connection is not guaranteed, due to its replication and synchronisation capabilities. The BBC uses it for its dynamic content platforms. It can be used in applications such as CRM and CMS, where data is changed occasionally, and versioning is crucial. Cloudant is an enterprise software company that provides an open source distributed database service based on the Apache CouchDB project.

Its features are:

- CouchDB provides ACID semantics by implementing a form of Multi-Version Concurrency Control (high volume of concurrent readers and writers without conflict).
- CouchDB supports bi-directional replication (or synchronisation) and offline operation.
- Unique URI that gets exposed via HTTP. REST uses the POST, GET, PUT and DELETE HTTP methods for the four CRUD operations.
- It assures eventual consistency (a model used in the domain of parallel programming) to be able to provide both availability and partition tolerance.

MongoDB

MongoDB is an open source, scalable, high-performance, and document-oriented database optimised for highly transient data, and is written in C++. It provides a RESTful API. A free cloud-based monitoring service is provided for MongoDB deployments. It supports search by range queries, fields and regular expressions. Master-slave replication is supported, where the master can perform read and write operations, while slaves can read or take back-ups. MongoDB supports horizontal scaling with the use of sharding. It can be effectively used as an efficient file storage, which is capable of taking the benefits of load balancing and data replication.

Use-cases:

- Flexible schemas are the best fit for document and content management systems.
- Good fit in conjunction with RDBMS, for e-commerce infrastructure.
- Good fit for gaming, due to high read-write performance.
- Very efficient for server-side infrastructure of mobile applications.

Graph databases

Graph databases store graph-oriented data with nodes—entities such as people, businesses, accounts and edges. They connect nodes to nodes, or nodes to properties, and represent the relationship and properties, where every element holds a direct pointer to its neighbouring